

Relief Generator

by FlorindoLAB

Height maps and displacement maps,
right in your browser.

User guide

Version 1.0 beta

florindolab.com

Overview

Relief Generator creates **height maps** and **displacement maps** for Blender, Unreal Engine, Unity, Godot, Maya, 3ds Max, Cinema 4D, Houdini, motion design and VFX. It runs entirely in your browser: nothing to install, no account, and **no image or project is ever sent to a server**.

WHAT THE TOOL DOES

- **60 presets**
  Mountains, volcanoes, dunes, islands, canyons, surfaces, patterns… with a new variation in one click.
- **Non-destructive layers**
  Generators, noises, images, filters, erosion and sculpting, stacked and combined.
- **Erosion**
  Hydraulic (rain, gullies) and thermal (scree), identical in preview and export.
- **Sculpting and masks**
  8 brushes in 2D and 3D, and painted masks on every layer.
- **Seamless tiling**
  Maps that repeat with no visible seam, checked automatically.
- **Complete exports**
  8- and 16-bit PNG, 16-bit RAW, 32-bit EXR and derived maps, up to 8192 px.

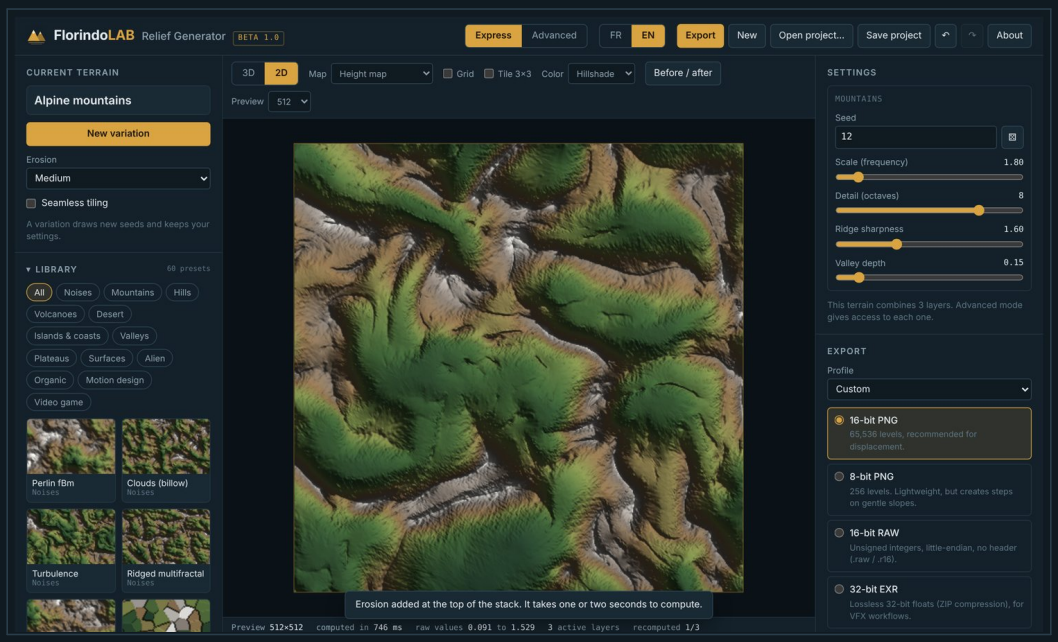
REQUIREMENTS

- A recent version of Chrome, Edge, Firefox or Safari, on a computer.
- WebGL for the 3D view (the 2D view works without it).
- 8 GB of memory recommended for 8192 × 8192 px exports.
- Interface in French and English (FR / EN buttons in the header).

Getting started (Express mode)

Express mode gets you a result in a few clicks. It is active when the tool opens.

1. Open the tool at **florindolab.com/tools/relief-generator/**.
2. In the **Library**, pick a terrain. The tags (Mountains, Volcanoes, Desert…) filter the list.
3. Adjust the few settings offered in the right panel.
4. **New variation** draws new seeds and keeps your settings. The **Erosion** menu adds ready-made erosion. **Seamless tiling** makes the map repeatable.
5. Check the result in **3D** or **2D** (tabs at the top of the view).
6. Click **Export** in the header: choose a profile, a format and a resolution, then **Export the height map**.



Express mode: current terrain, quick erosion and the library on the left, settings and export on the right.

> **Tip** · Click the library title to collapse it and keep erosion and global settings within reach.

The interface

Area	Contents
Header	Express / Advanced mode, FR / EN language, Export, New (empty project), Open and Save project, Undo / Redo, About.
Left panel	The preset library (Express) or the layer stack (Advanced), your custom presets and the global settings.
Center view	3D or 2D view, displayed map, display options, Before / after, preview resolution.
Right panel	Settings of the selected layer, export, derived maps.
Status bar	Preview size, computation time, recomputed layers, value of the hovered pixel (2D), seam check.

MOVING AROUND THE VIEW

	3D view	2D view
Left click	Orbit around the terrain	Drag to pan
Right click	Pan the view	Pan the view
Mouse wheel	Zoom	Zoom around the cursor
Double-click		Reframe the map
While sculpting	Left click paints, right click orbits	Left click paints, right click pans

DISPLAY OPTIONS

- **Height** and **Light** (3D): vertical exaggeration and lighting direction.
- **Color**: Elevation, Grayscale, Hillshade or Clay. **Wireframe** shows the triangles.
- **Grid** and **Tile 3×3** (2D): framing guides and a tiling preview.
- **Before / after**: hold the button to see the state before your last change.
- **Preview** 128, 256 or 512: preview detail. Export is always computed at the chosen resolution.

> **Tip** · Display settings never change the exported map. Preview and export use the same coordinates: only the level of detail changes.

Advanced mode: layers

The terrain is a **stack of layers**. The bottom layer lays down the relief (it is marked **base**), then each layer above is combined with the result. The top layer is applied last.

Badge	Layer type	Role
G	Generator	Creates relief: noises, mountains, dunes, volcano, shapes…
IM	Image	Turns an image into relief (levels, contrast, gamma, smoothing…).
FX	Filter	Changes the result of the layers below: smooth, terraces, curve, levels, invert.
ER	Erosion	Simulates hydraulic or thermal erosion on the layers below.
SC	Sculpt	Holds your brush strokes, replayed on every computation.

MANAGING THE STACK

- **Add**: pick the type in the menu below the stack, then “+ Add”. You can also drop an image on the view.
- **Reorder**: drag a layer by its handle (the dots on the left of the row), or use the ↑ ↓ arrows.
- **Delete**: the cross on the row, or the Delete key. **Duplicate** and rename: in the right panel.
- **Enable / disable**: the checkbox on each row. **Solo layer** (view bar) isolates the selected layer.

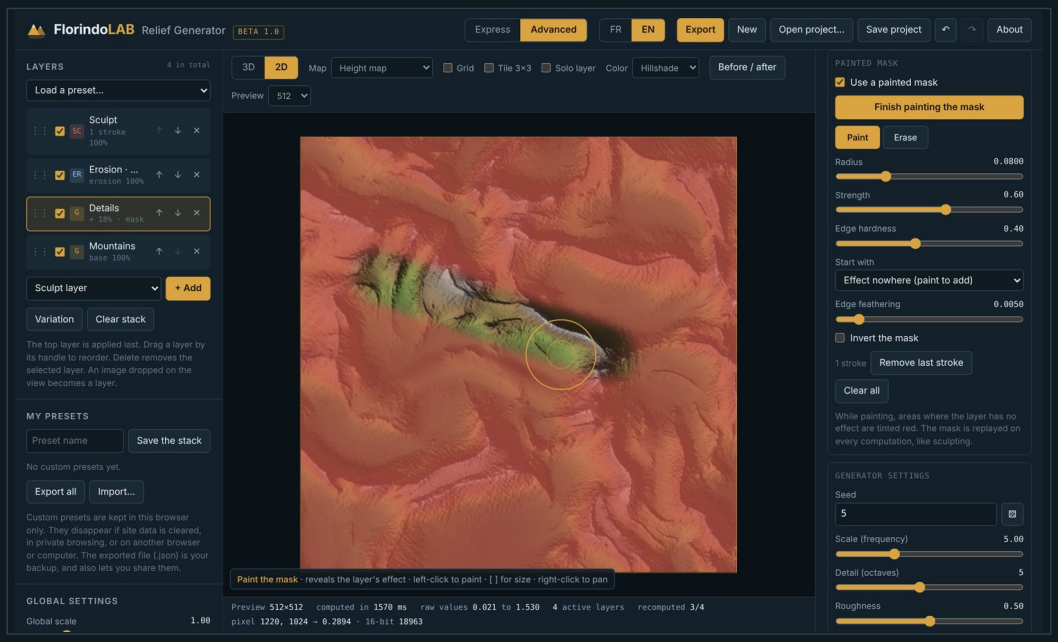
BLEND MODES

Mode	Effect
Normal	Replaces the relief below, according to the intensity.
Add / Subtract	Adds the layer to the relief below, or removes it: ideal for details.
Multiply	Modulates the relief below (dark areas of the layer = flattened relief).
Maximum / Minimum	Keeps the higher or the lower of the two reliefs.
Mask	The layer acts as a mask for the relief below.

Each layer can also be **limited by elevation** (it only acts between two heights) and get a **painted mask**.

PAINTED MASKS

1. Select the layer, then tick **Use a painted mask** (right panel).
2. Choose **Start with**: “effect everywhere” (you paint to remove) or “effect nowhere” (you paint to add).
3. Click **Paint the mask**, then paint in the view with **Paint** or **Erase**. Areas where the layer has no effect are tinted red.
4. Adjust the **edge feathering** if needed, then click **Finish painting the mask**.



Painting a mask: the “Details” layer only acts inside the painted area.

IMPORTING AN IMAGE

Choose “Imported image…” in the add menu, or drop a PNG, JPEG or WebP file on the view. Then adjust framing, black and white points, contrast, brightness, gamma, smoothing and sharpness.

> **Tip** · Browsers read images as 8-bit, even 16-bit PNGs. To keep full precision, prefer terrain generated in the tool.

Generators and noises

Generator	Use
Fractal noise	Perlin, Simplex or Value, as fBm, Billow, Ridged or Turbulence. The base of most terrains.
Voronoi	Cells, cell edges, F1 / F2 distances, flat or domed cells. Cobblestones, scales, cracks.
Hills, Mountains	Natural terrain. Mountains: ridge sharpness and valley depth.
Dunes	Wind direction, asymmetry, undulation and sand ripples.
Craters, Volcano	Crater fields, volcano with crater and ravines.
Canyons, Plateaus & terraces	Winding gorges with strata, regular steps.
Islands & archipelagos	Sea level, gathering toward the center, coast steepness.
Rocky surface	Blocks, cracks and grain, for surface displacement.

Generator	Use
Organic relief, Abstract patterns	Twisted shapes, ripples, interference, spirals, stripes.
Primitive shape	Dome, cone, pyramid, ring, plateau, slope: to guide a terrain.

COMMON SETTINGS

- **Seed**: the same number gives exactly the same terrain. The die button draws a random one.
- **Scale**: size of the shapes. **Detail (octaves)**: number of stacked detail levels.
- **Roughness** and **Lacunarity**: weight and fineness of small details. **Warp**: twists the shapes.
- **Offset X / Y**: slides the terrain without changing it.

> **Tip** · Double-click a slider to reset it to its default value.

GLOBAL SETTINGS

Global scale enlarges or shrinks the whole terrain. **Height curve** deepens valleys or swells peaks. **Normalize** spreads the terrain over the full 0 to 1 range. **Invert** flips the terrain. **Seamless tiling** makes the map repeatable.

Erosion

Add a **Hydraulic erosion** or **Thermal erosion** layer above the layers to erode. In Express mode, the **Erosion** menu does the same with a preset (Light, Medium, Strong, Gullied, Scree).

- **Hydraulic**: raindrops dig gullies and deposit sediment lower down. Main settings: rain amount, erosion strength, deposition, gully width.
- **Thermal**: material slides down wherever the slope exceeds the talus angle. Softens steep slopes and forms scree.
- **Simulation resolution** (256, 512, 1024): erosion is simulated on a fixed grid, then applied to the map, so preview and export give the same result. 1024 gives finer gullies but takes a few seconds.

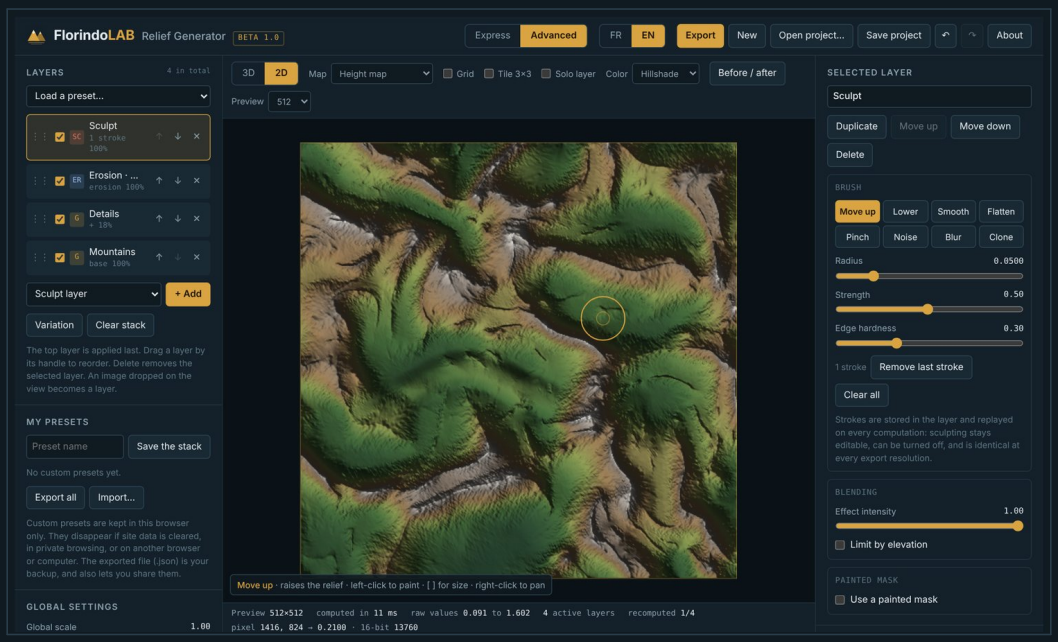
> **Tip** · Put small details (fine noise, sculpting) above the erosion: they stay crisp.

Sculpting

Add a **Sculpt layer**, select it and paint directly in the 3D or 2D view. Strokes are stored in the layer and replayed on every computation: sculpting stays editable and identical at every export resolution.

Brush	Effect
Raise / Lower	Adds or removes material.
Smooth / Blur	Softens small details, or strongly.
Flatten	Brings back to the height where the stroke starts.
Pinch	Sharpens ridges.
Noise	Adds irregularities.
Clone	Copies an area: Alt+click to pick the source, then paint elsewhere.

- **Radius**, **Strength**, **Edge hardness** in the right panel. The **[** and **]** keys change the size.
- **Remove last stroke**, **Clear all**, or Ctrl+Z.

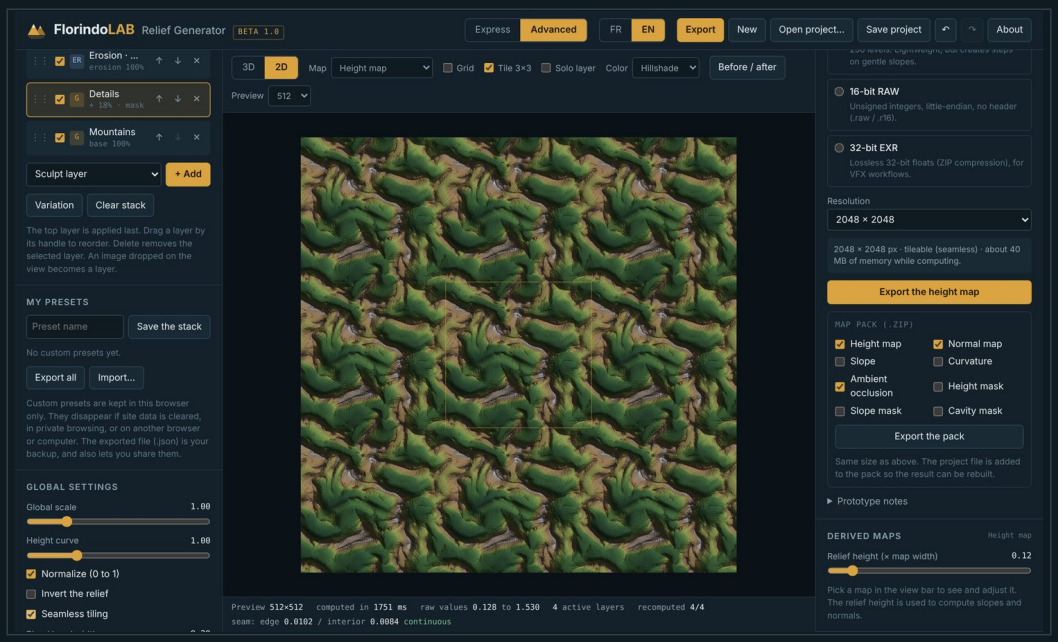


A sculpt layer: the yellow circle shows the brush, its settings are on the right.

> **Tip** · With “Normalize” ticked, a peak sculpted higher than the rest rescales the whole map. Untick it while sculpting to keep your heights.

Seamless tiling

Tick **Seamless tiling** (Current terrain block in Express, or global settings). The whole stack tiles: generators, images, filters, erosion, sculpting, masks and derived maps. The 2D view switches to **Tile 3×3** so you can check the result, and the status bar shows “continuous” when the edges match. The **blend band width** sets how far from the edges the terrain is blended.



A seamless map tiled 3 × 3: no visible seam.

Derived maps

The **Map** menu (view bar) shows maps computed from the final height map. Their settings are at the bottom of the right panel. The **relief height** (in map widths) is used to compute slopes and normals.

Map	Use
Normal map	Lighting detail in a shader. OpenGL convention (Blender, Unity, Godot) or DirectX (Unreal).
Slope, Curvature	Material masks: rock on slopes, wear on edges.
Ambient occlusion	Darkens crevices, for texturing or compositing.
Height, slope and cavity masks	Black and white, to distribute snow, grass, soil…

Export

The **Export** button in the header takes you to the Export section. Export recomputes the terrain at the chosen resolution, with a progress bar.

Format	What for
16-bit PNG	The default: 65,536 levels, no visible steps in displacement.
8-bit PNG	Lightweight, but creates steps on gentle slopes.
16-bit RAW	Unsigned integers, little-endian, no header (.raw / .r16): Unity and Unreal terrains.
32-bit EXR	Lossless floats (ZIP compression): VFX, Nuke, Houdini, Maya, Blender.

Profile	Settings and import
Blender	16-bit PNG, 4096 px. Displace modifier or Displacement node, image in “Non-Color” space.
Unreal Engine	16-bit PNG, 2017 px. Landscape › Import from File. Recommended sizes: 1009, 2017, 4033 or 8129 px.
Unity	16-bit RAW, 2049 px. Terrain › Import Raw, Depth 16 bit, Byte Order Windows. Size $2^n + 1$.
Godot	16-bit PNG, 2048 px. Terrain3D or HTerrain; height scale is set in the plugin.
Maya	32-bit EXR, 4096 px. File node in “Raw” color space with “Alpha is Luminance” enabled, into a displacementShader. With Arnold: mesh subdivision, height in its Displacement attributes.
3ds Max	16-bit PNG, 4096 px. Displace modifier (Bitmap) or the renderer's displacement. When loading the image: “Override” with a gamma of 1.0.
Cinema 4D	16-bit PNG, 4096 px. Displacer deformer (Shader › Bitmap) or Displacement channel with Sub-Polygon Displacement. Bitmap color profile: “Linear”.
Houdini	32-bit EXR, 2048 px. “HeightField File” SOP for a terrain, or a displacement texture in Karma / Mantra without gamma correction.

- **Custom size**: from 16 to 8192 px, square or rectangular.
- **Map pack (.zip)**: tick the maps you want (height, normal, occlusion…) then “Export the pack”. The project file is added to the pack.

- Files are named automatically: terrain, size, “_seamless” when relevant, bit depth.



The Export section with the Unreal Engine profile.

Projects and presets

- **Save project** creates a .fhmap.json file: layers, seeds, settings, export and imported images. **Open project…** reloads it exactly.
- **New** starts from an empty stack. Ctrl+Z brings the previous project back.
- **My presets** (Advanced mode): save the current stack as a preset, then reload it in one click. **Export all** and **Import…** let you back them up and share them.

> **Tip** · Custom presets are kept in your browser only. They disappear if site data is cleared, in private browsing, or on another computer. A banner reminds you to export the ones that never were.

Shortcuts

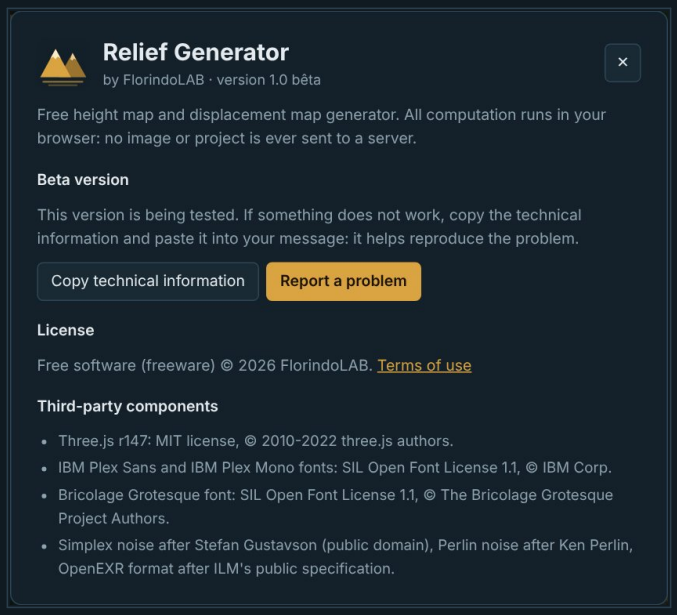
Shortcut	Action
Ctrl+Z	Undo
Ctrl+Y or Ctrl+Shift+Z	Redo
Delete	Delete the selected layer (Advanced mode)
[and]	Shrink or enlarge the brush (sculpting, mask)
Alt+click	Pick the clone source
Double-click a slider	Default value
Double-click in the 2D view	Reframe the map

Troubleshooting

Problem	Solution
The 3D view stays empty	The browser has no WebGL or has it disabled. Update it or try another browser; the 2D view remains available.
Export is very slow or fails	Lower the resolution or close other tabs. At 8192 px, expect a few seconds to over a minute.
My presets are gone	Site data was cleared, or you were in private browsing. Import your exported file again.
The terrain moves when I sculpt	Normalization rescales the map. Untick “Normalize” in the global settings.
A 16-bit image looks stepped	The browser reads it as 8-bit. Add a little smoothing to the image layer.

REPORTING A PROBLEM

This is a beta version: your feedback matters. Open **About**, click **Copy technical information**, then **Report a problem** and paste it into your message (florindolab.com/contact/).



The About window.

LICENSE AND PRIVACY

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